

JOSHUA DROSSMAN

☎ (609) 751-1349 ✉ jddrossman@gmail.com [in linkedin.com/in/joshua-drossman](https://www.linkedin.com/in/joshua-drossman) github.com/joshddrossman

Education

Massachusetts Institute of Technology

Sep. 2024 – May 2028 (expected)

Ph.D. in Operations Research, GPA 5.0/5.0. Advisor: Prof. Alexandre Jacquillat

Cambridge, MA

- Coursework: Machine Learning with Optimization, Probability Theory, Multimodal AI.

Princeton University

Sep. 2018 – May 2022

B.S.E. in Operations Research and Financial Engineering, magna cum laude, GPA 3.92/4.0

Princeton, NJ

- Coursework: Regression & Applied Time Series, Econometrics, Analysis of Big Data.
- Senior thesis: *Managing Uncertainties in the Development of CO₂ Capture, Transport, & Storage Infrastructure: A Scenario Optimization Approach*; network optimization of CO₂ pipeline deployment under uncertain capital costs.

Research Experience

MIT Operations Research Center

Sep. 2024 – Present

Graduate Research Assistant

Cambridge, MA

- **Experimental evaluation of LLM agents.** Designed a replicable and scalable methodology to evaluate LLM agents used for interactive combinatorial optimization and decision support.
- Designed agents in LangGraph that call optimization solvers through MCP tool servers, varying tool access, prompt engineering, and solver support across 14 designs; built the Python harness to evaluate them and conduct statistical tests.
- **Large-scale optimization for election planning.** Built MIP models for polling-place location, voter assignment, and contiguous precinct redistricting for five Montana counties; solutions reduced average voter travel burden by 5-12%.
- Developed an exact decomposition method for hierarchical districting and facility location: logic-based Benders with an NP-hard subproblem, accelerated by a spanning-tree DP oracle and a new family of problem-specific cuts; solves instances 100x larger than previous methods in the literature.
- Prototyped an interactive planning tool for election officials to explore and visualize optimized plans and generate their own.

Princeton University

May 2020 – May 2022

Undergraduate Research Assistant

Princeton, NJ

- Performed sensitivity analysis with and refined a linear programming capacity expansion model that co-optimizes investment and dispatch across electricity, fuels, and end-use demand over multi-decade horizons.

Industry Experience

Wood Mackenzie

Aug. 2022 – Aug. 2024

Supply Chain Analyst, promoted to Senior Analyst

Boston, MA

- Built regression and time-series forecasting models on large-scale client transaction data and market indicators, producing price-benchmark distributions, cost models, and forecasts that informed client procurement and investment decisions.
- Built Python tools to collect, clean, and mine a proprietary transaction database, using NLP on unstructured records; benchmarked client spend against market trends to identify multi-million-dollar savings.
- Owned analytics delivery for utility, oil & gas, and natural-resources clients, presenting findings to procurement and finance leadership and translating model outputs into recommendations; promoted after two years.

Publications and Working Papers

J. Drossman, A. Jacquillat, S. Martin. “Let’s Have a Conversation: Designing and Evaluating LLM Agents for Interactive Optimization.” Under review at *Manufacturing & Service Operations Management*, 2026. arXiv preprint.

J. Drossman, A. Jacquillat, B. Sturt. “Exact Decomposition for Hierarchical Precinct Districting.” Working paper, 2026.

J. Drossman, A. Jacquillat, C. Podimata. “Predict-then-Optimize for Sterilization and Rabies Control in India.” In prep.

Skills, Teaching, and Talks

Languages: Python, SQL, Julia, Java

Statistics: Time-series forecasting, regression, statistical inference, hypothesis testing, experimental design

ML: Deep learning, reinforcement learning, fine-tuning foundation models, NLP; PyTorch, scikit-learn, pandas, NumPy

Data / Visualization: Tableau, SQL, matplotlib, seaborn, folium, GeoPandas

Optimization: Gurobi, CPLEX, JuMP; integer, robust, and stochastic optimization

Teaching: 15.C57 Optimization Methods (MIT, Fall 2025); 15.071 The AI Edge (MIT, Fall 2026)

Talks: “Designing and Evaluating LLM Agents for Interactive Optimization,” Michigan Ross AI Workshop, May 2025